SIGNALS AND SYSTEMS

Signals, Systems and Linear Time-Invariant Systems

Lecture notes for the first part of the course: what a signal is, how systems are classified, and why linear time-invariant systems are described by a single function.

Covers	Level	Assumed background
Chapters 1 to 3	Undergraduate, second year	Calculus, complex numbers, basic circuits

How to read these notes

Each chapter builds on the one before it. Within a chapter, every idea is introduced in the same order: a picture, a definition, an equation, a short derivation, a worked example, and a warning about the mistake that is easiest to make. Worked examples always use five headings: Given, Find, Method, Solution, Check. Do the Check step yourself before reading it.

Two conventions apply everywhere. First, energy and power are **normalised**: the resistance is taken as $1\ \Omega$, so instantaneous power is $|x|^2$. Second, the imaginary unit is written j.

CHAPTER 1

Signals

A signal is information written as a function. This chapter says what that means, how much energy or power a signal carries, how it changes when we shift or stretch the time axis, and when it repeats.

1.1 What a signal is

A signal is a physical quantity that varies and carries information. Mathematically, it is a function of one or more independent variables.

In this course the independent variable is almost always time. It does not have to be. An image is a signal of two space variables, and the same algebra applies.

There are two kinds of signal, and they are kept apart throughout the course.

CONTINUOUS TIME

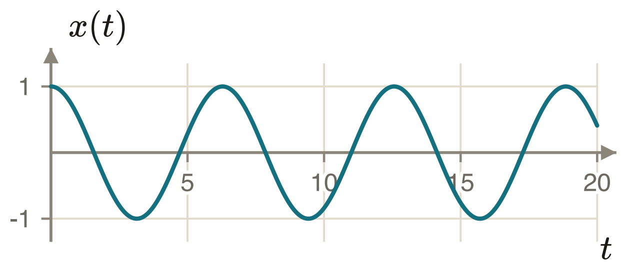
$$x(t), \qquad t \in \mathbb{R}$$

Round brackets. The signal has a value at every real instant.

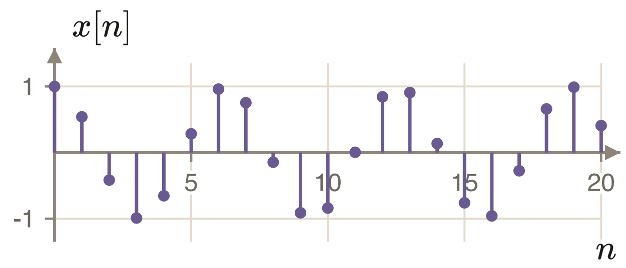
DISCRETE TIME

$$x[n], \qquad n \in \mathbb{Z}$$

Square brackets. The signal has a value only at integer indices. Here n is an integer **index**, not a time in seconds.



A continuous-time signal is drawn as an unbroken curve.



A discrete-time signal is drawn with stems. The signal is the dots. There is nothing between them.

NOTATION

Writing $x[t]$ or $x(n)$ is not a small slip. It states the wrong domain, and every later statement about periodicity, convolution limits and transforms then breaks.

1.2 Signal energy and power

Start with a resistor. If $v(t)$ is the voltage across a resistance R, the instantaneous power is

$$p(t) = v(t)\, i(t) = v(t) \left(\frac{v(t)}{R} \right) = \frac{1}{R} v^2(t).$$

Energy is power multiplied by time. Because the power varies, we integrate:

$$E = \int_{t_1}^{t_2} p(t)\, \mathrm{d}t = \int_{t_1}^{t_2} \frac{1}{R} v^2(t)\, \mathrm{d}t.$$

From here on we set $R = 1\ \Omega$ and work with $|x(t)|^2$. This is the normalised convention. Put the factor $1/R$ back whenever a physical number in joules or watts is required.

We use the modulus because a signal may be complex valued, and $|x(t)|^2 = x(t)\, x^*(t)$ is the quantity that is real and never negative. Writing $x^2(t)$ instead is correct only for real signals.

Total energy

TOTAL ENERGY OVER ALL TIME

$$E_\infty = \lim_{T\to\infty} \int_{-T}^{T} |x(t)|^2 \,\mathrm{d}t = \int_{-\infty}^{\infty} |x(t)|^2 \,\mathrm{d}t$$

$$E_\infty = \lim_{N\to\infty} \sum_{n=-N}^{N} |x[n]|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

The integral or the sum may not converge. E_∞ is defined as a limit, and a limit that fails to exist is not a large number. It is the absence of an answer, and that fact is useful.

Average power

If the energy is infinite, ask instead how fast energy arrives. Divide by the length of the window and let the window grow.

AVERAGE POWER OVER ALL TIME

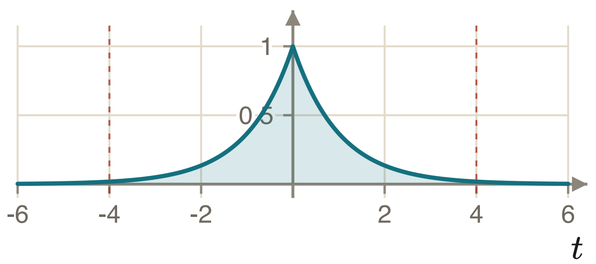
$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \,\mathrm{d}t$$

$$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

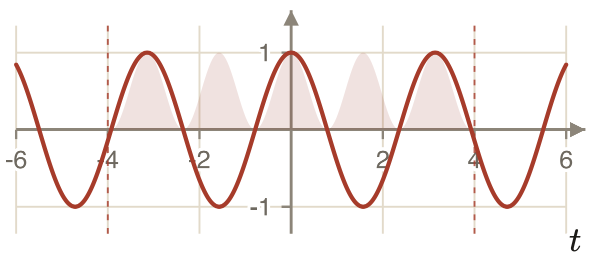
In discrete time the divisor is $2N+1$, because that is the number of samples between $-N$ and $+N$ inclusive.

READ THE DEFINITION CAREFULLY

P_∞ is not energy divided by infinity. It is the limit of a ratio, and that ratio can settle on any value, including zero.



Converging. As the window grows, the shaded area approaches a finite limit.



Diverging. Every period adds the same area, so $E_\infty \to \infty$ and only the average power is meaningful.

1.3 Energy signals, power signals, and neither

TYPE	CONDITION	TYPICAL SIGNALS
Energy signal	$E_\infty < \infty$ and $P_\infty = 0$	Pulses, decaying responses, anything that ends
Power signal	$E_\infty \to \infty$ and $0 < P_\infty < \infty$	Constants, sinusoids, signals that run forever
Neither	$E_\infty \to \infty$ and $P_\infty \to \infty$	Signals that grow without bound, such as $t\,u(t)$

The two conditions in each row are not independent. Finite energy forces $P_\infty = 0$, because a finite number divided by $2T \to \infty$ goes to zero. The second condition is a consequence, not a second test.

EXAMPLE 1.1

GIVEN $x(t) = 1$ for $0 \le t \le 1$, and $x(t) = 0$ otherwise.

FIND Is this an energy signal or a power signal?

METHOD Find E_∞ first. If it is finite, the classification is settled.

SOLUTION

$$E_\infty = \int_0^1 1^2 \, \mathrm{d}t = 1 < \infty$$

$$P_\infty = \lim_{T\to\infty} \frac{1}{2T} \int_0^1 1 \, \mathrm{d}t = \lim_{T\to\infty} \frac{1}{2T} = 0$$

So $x(t)$ is an **energy signal**, with $E_\infty = 1$ J.

CHECK Halve the amplitude. The energy must fall by a factor of four, and $\int_0^1 (1/2)^2 \mathrm{d}t = 1/4$. The dependence is quadratic, as it must be for a squared quantity.

EXAMPLE 1.2

GIVEN $x[n] = 4$ for every integer n.

FIND Is this an energy signal or a power signal?

METHOD The energy clearly diverges, so go straight to the average power and count the samples correctly.

SOLUTION

$$E_\infty = \sum_{n=-\infty}^{\infty} |4|^2 \to \infty$$

$$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} 16 = \lim_{N\to\infty} \frac{(2N+1)\cdot 16}{2N+1} = 16$$

So $x[n]$ is a **power signal**, with $P_\infty = 16$.

CHECK A constant of amplitude A must have $P_\infty = A^2$. The factor $2N+1$ cancels exactly, which confirms the sample count was right.

COMMON MISTAKE

Concluding that infinite energy means infinite power. The factor $1/(2N+1)$ does the work. A sum that diverges, divided by a count that also diverges, can converge.

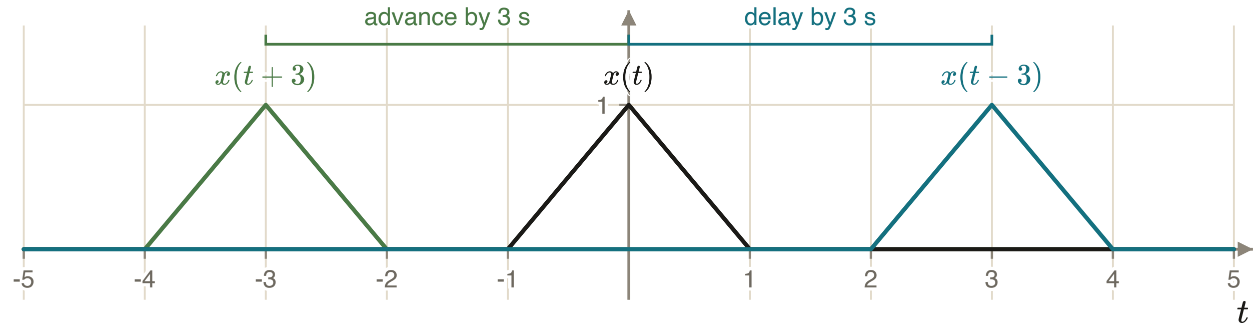
1.4 Time shifting

TIME SHIFT

$$x(t) \;\longrightarrow\; x(t - t_0)$$

If $t_0 > 0$ the signal is **delayed** and moves right. If $t_0 < 0$ it is **advanced** and moves left.

The sign feels backwards until you read it as a question about when. The output at time t shows what the input was doing at the earlier time $t - t_0$. The value has been held back by t_0 seconds.



Shifting moves the signal along the time axis. The shape is untouched.

Discrete time is identical: $x[n] \to x[n-n_0]$ with n_0 an integer. A one-sample delay, $x[n-1]$, is the basic memory element of every difference equation in Chapter 3.

1.5 Time reversal and time scaling

TIME REVERSAL

$$x(t) \longrightarrow x(-t) \qquad (x[n] \to x[-n])$$

A flip about the vertical axis.

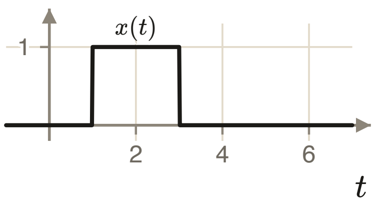
TIME SCALING

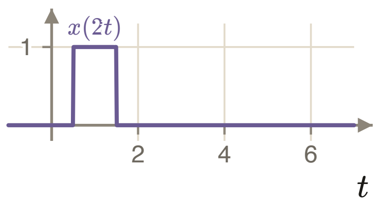
$$y(t) = x(at), \qquad a > 0$$

If $a > 1$ the signal is compressed and speeded up. If $0 < a < 1$ it is stretched and slowed down.

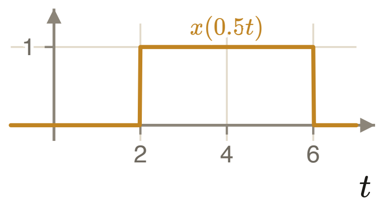
Write the scaled signal as a new signal $y(t)$. Writing $x(t) = x(at)$ would force $a = 1$.

Scaling changes the support in a predictable way. If x is non-zero only on $[\alpha, \beta]$, then $x(at)$ is non-zero only on $[\alpha/a, \beta/a]$. The width is divided by a.





Compressed: support $[0.5, 1.5]$.



Stretched: support $[2, 6]$.

DISCRETE TIME IS NOT SYMMETRIC

$x[2n]$ throws away the odd-indexed samples. It is real decimation and it destroys information. And $x[n/2]$ is not defined at odd n without an extra rule for filling the gaps. Continuous-time scaling can be undone; discrete-time decimation cannot.

1.6 Combining a shift and a scale

To find $x(at-b)$, do the two operations in this order.

SHIFT, THEN SCALE

$$(1) \quad v(t) = x(t-b) \qquad\qquad (2) \quad y(t) = v(at) = x(at-b)$$

Shift by b first. Then scale the result by a.

WHY THE ORDER MATTERS

Scaling first gives $w(t) = x(at)$. Shifting that by b gives $w(t-b) = x\big(a(t-b)\big) = x(at-ab)$. Unless $a = 1$ this is a different signal: it is shifted by b instead of by b/a.

EXAMPLE 1.3

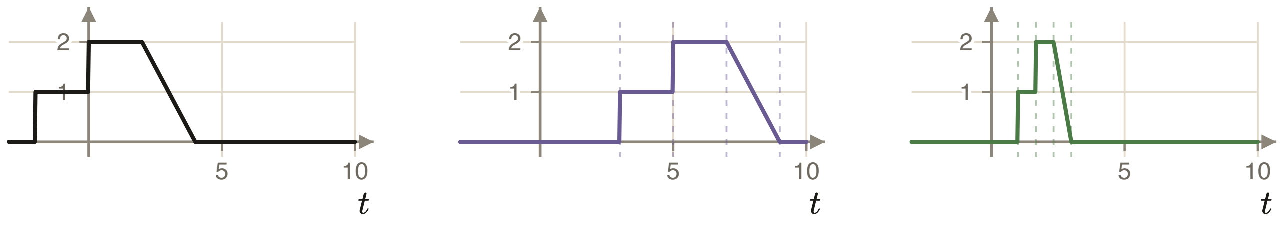
GIVEN $x(t)$ is zero for $t < -2$, equal to 1 on $[-2, 0]$, equal to 2 on $[0, 2]$, and falls linearly from 2 to 0 on $[2, 4]$.

FIND Plot $x(3t-5)$.

METHOD Here $a = 3$ and $b = 5$. Shift right by 5, then compress by 3.

SOLUTION $v(t) = x(t-5)$ has corners at $t = 3, 5, 7, 9$. Then $y(t) = v(3t)$ has corners at $t = 1,\ 5/3,\ 7/3,\ 3$.

CHECK The total duration must shrink by exactly $a = 3$. The original spans $[-2, 4]$, width 6. The result spans $[1, 3]$, width 2. Also, y reproduces $x(0)$ where $at - b = 0$, that is at $t = b/a = 5/3$, which is where the level 2 begins.



$x(t)$ $v(t) = x(t-5)$ $y(t) = x(3t-5)$

1.7 Periodic signals

PERIODICITY

$$x(t) = x(t+T) \quad \text{for all } t \in \mathbb{R}, \text{ for some } T > 0$$

$$x[n] = x[n+N] \quad \text{for all } n \in \mathbb{Z}, \text{ for some integer } N > 0$$

A signal that is not periodic is **aperiodic**.

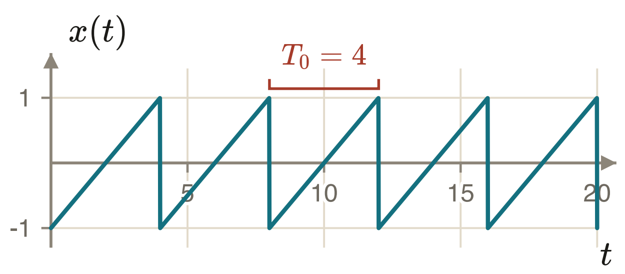
Note both quantifiers. There must **exist** a period, and it must work for **all** t or n. In continuous time T may be any positive real number. In discrete time N must be an integer. There is no period of 3.5 samples. Almost every discrete-time surprise in this course comes from that one line.

If T is a period then so is $2T$, $3T$, and so on. The **fundamental period** T_0 is the smallest positive period. The same applies to N_0 in discrete time.

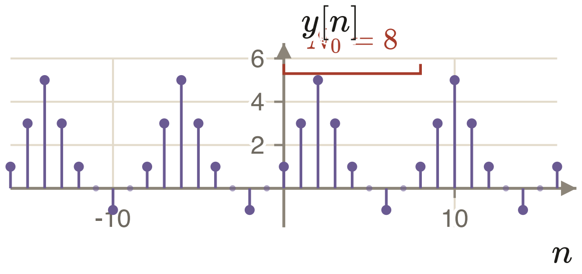
FUNDAMENTAL FREQUENCY

$$\omega_0 = \frac{2\pi}{T_0} \qquad \text{and} \qquad \omega_0 = \frac{2\pi}{N_0}$$

Without the word *smallest*, ω_0 would not be well defined.



Periods are $4, 8, 12, \ldots$ The fundamental period is $T_0 = 4$.



Periods are $8, 16, 24, \ldots$ The fundamental period is $N_0 = 8$.

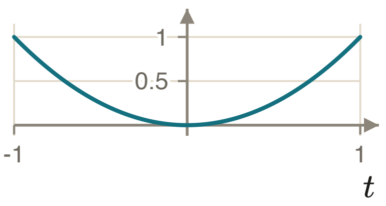
1.8 Even and odd signals

A signal is **even** if $x(t) = x(-t)$, and **odd** if $x(t) = -x(-t)$. The same definitions apply to $x[n]$ with n in place of t.

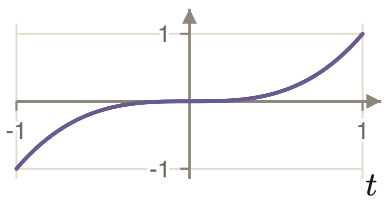
Setting $t = 0$ in the odd condition gives $x(0) = -x(0)$, so $x(0) = 0$. A signal with $x(0) \neq 0$ cannot be odd. That is a one-second test.

Every signal splits into an even part and an odd part, in exactly one way:

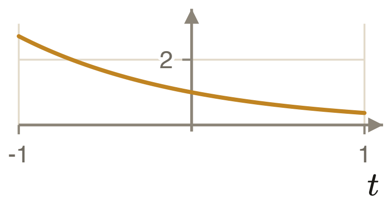
> EVEN AND ODD PARTS
>
> $$\mathcal{E}\text{v}\{x(t)\} = \tfrac{1}{2}\big[x(t) + x(-t)\big], \qquad \mathcal{O}\text{d}\{x(t)\} = \tfrac{1}{2}\big[x(t) - x(-t)\big]$$
>
> $$x(t) = \mathcal{E}\text{v}\{x(t)\} + \mathcal{O}\text{d}\{x(t)\}$$
>
> Adding the two definitions returns x exactly.



t^2 is even.



t^3 is odd.



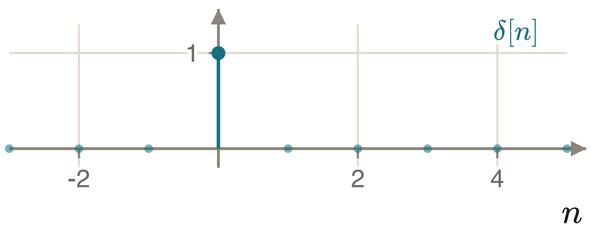
e^{-t} is neither.

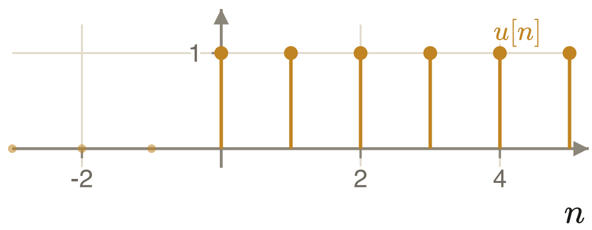
1.9 The unit impulse and the unit step in discrete time

> DEFINITIONS
>
> $$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$
>
> $$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$
>
> Both are ordinary sequences. Nothing infinite happens here.

They are related by a difference and by a running sum. These are the discrete versions of differentiation and integration.

> FIRST DIFFERENCE AND RUNNING SUM
>
> $$\delta[n] = u[n] - u[n-1]$$
>
> $$u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \delta[n] + \delta[n-1] + \delta[n-2] + \cdots$$





Sampling and sifting

Two properties do most of the work later. They say different things, and the difference matters.

SAMPLING PROPERTY

$$x[n]\,\delta[n-n_0] = x[n_0]\,\delta[n-n_0]$$

Both sides are **sequences**. Multiplying by a shifted impulse keeps one sample and deletes the rest.

SIFTING PROPERTY

$$x[n_0] = \sum_{n=-\infty}^{\infty} x[n]\,\delta[n-n_0]$$

The right-hand side is a **number**. Sifting is sampling followed by a sum.

EXAMPLE 1.4

GIVEN $x[0]=1$, $x[1]=2$, $x[2]=3$, and $x[n]=0$ elsewhere. Take $n_0=2$.

FIND The sampled sequence and the sifted value.

SOLUTION Sampling: $x[n]\delta[n-2]=x[2]\delta[n-2]=3\,\delta[n-2]$, a single stem of height 3 at $n=2$.
Sifting: $\sum_n x[n]\delta[n-2]=x[0]\cdot 0+x[1]\cdot 0+x[2]\cdot 1=3$.

CHECK The first answer is a signal. The second is a number. If your two answers have the same type, one of them is wrong.

Applying the sampling property at every shift and adding the results gives the identity that Chapter 3 is built on.

REPRESENTATION PROPERTY

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

Any sequence is a sum of weighted, shifted impulses. The weights are the sample values themselves.

1.10 The unit impulse and the unit step in continuous time

The continuous-time step is straightforward: $u(t)=1$ for $t\geq 0$ and 0 otherwise. The impulse needs more care.

THE IMPULSE IS NOT A FUNCTION

It is often written as $\delta(t)=\infty$ at $t=0$ and 0 elsewhere, with total area 1. No ordinary function behaves like that: a function that is zero except at one point has integral zero. δ is a **generalized function**, or distribution. It is defined by what it does inside an integral, not by its values.

DEFINING PROPERTY (SIFTING)

$$x(t_0) = \int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,\mathrm{d}t$$

This is the definition. Everything else about δ follows from it.

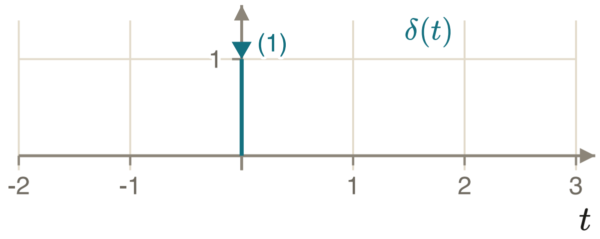
A useful picture: take a rectangle of width ε and height $1/\varepsilon$, so its area is 1 for every ε, and let $\varepsilon\to 0$. The limit does not exist point by point. It exists only under an integral sign, which is exactly the statement above.

The impulse and the step are related by differentiation and integration:

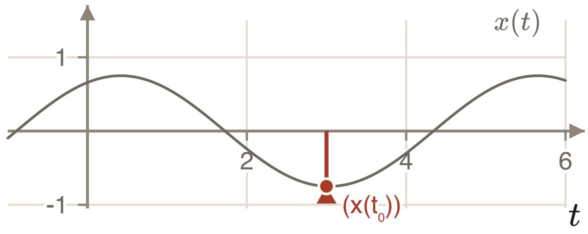
$$\delta(t) = \frac{\mathrm{d}}{\mathrm{d}t}u(t), \qquad u(t) = \int_{-\infty}^{t} \delta(\tau)\,\mathrm{d}\tau.$$

SAMPLING PROPERTY, CONTINUOUS TIME

$$x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$$



The arrow height shows the **area**, never a value of a function.



Sifting: the impulse at t_0 is scaled by the value of x there, and integrating returns that number.

1.11 Complex exponentials in continuous time

DEFINITION

$$x(t) = C\, e^{at}, \qquad C, a \in \mathbb{C}$$

The behaviour depends only on where a sits in the complex plane. There are three cases.

Case 1: C and a real

If $a < 0$ the signal decays. If $a > 0$ it grows. If $a = 0$ it is the constant C. A larger $|a|$ makes the decay or growth faster.

Case 2: $a = j\omega_0$ purely imaginary

Write $C = Ae^{j\theta}$ and use Euler's relation $e^{jx} = \cos x + j \sin x$:

$$x(t) = Ae^{j(\omega_0 t+\theta)} = A\cos(\omega_0 t + \theta) + jA\sin(\omega_0 t + \theta).$$

Here A is the amplitude, ω_0 is the angular frequency in rad/s, and θ is the phase in radians.

Is this signal periodic? Require $x(t) = x(t + T)$:

$$Ae^{j(\omega_0 t+\theta)} = Ae^{j(\omega_0 (t+T)+\theta)} \implies 1 = e^{j\omega_0 T} \implies j2\pi k = j\omega_0 T \implies T = \frac{2\pi}{\omega_0}k.$$

RESULT

Taking $k = 1$ gives the fundamental period $T_0 = 2\pi/\omega_0$. Every continuous-time complex exponential with $\omega_0 \neq 0$ is periodic. There is no extra condition.

EXAMPLE 1.5

GIVEN $x(t) = e^{j0.5\pi t}$.

FIND The fundamental period.

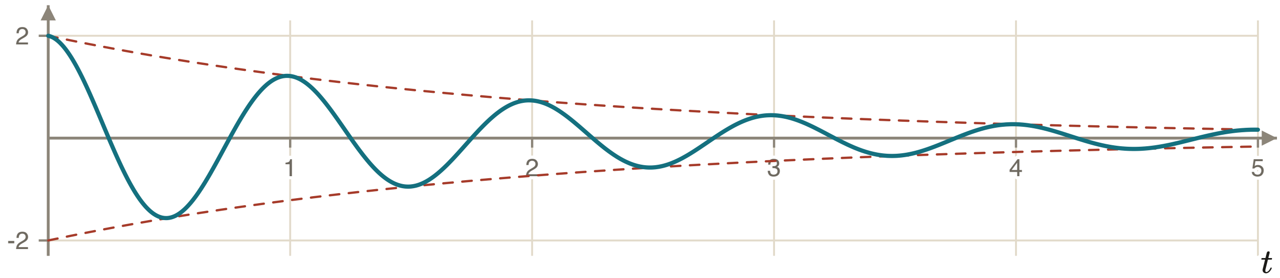
SOLUTION $T_0 = \dfrac{2\pi}{\omega_0} = \dfrac{2\pi}{0.5\pi} = 4$ seconds.

CHECK $0.5\pi \times 4 = 2\pi$, which is exactly one full turn of the phasor.

Case 3: $a = r + j\omega_0$

$$x(t) = Ae^{rt}\cos(\omega_0 t + \theta) + jAe^{rt}\sin(\omega_0 t + \theta).$$

This is a sinusoid inside an envelope $\pm Ae^{rt}$. If $r < 0$ the oscillation is damped, if $r > 0$ it grows, and if $r = 0$ it is sustained. This is the natural response of every second-order circuit.



A damped sinusoid, $\mathrm{Re}\{2e^{-0.5t}e^{j2\pi t}\}$, with its envelope shown dashed.

1.12 Complex exponentials in discrete time

DEFINITION

$$x[n] = C\,e^{\beta n} = C\,\alpha^n, \qquad \alpha = e^{\beta}$$

The power form is the natural one in discrete time. It is what a difference equation produces.

For real C and α: if $0 < \alpha < 1$ the sequence decreases; if $\alpha > 1$ it increases. For complex $C = |C|e^{j\theta}$ and $\alpha = |\alpha|e^{j\omega_0}$:

$$x[n] = |C|\,|\alpha|^n \cos(\omega_0 n + \theta) + j\,|C|\,|\alpha|^n \sin(\omega_0 n + \theta).$$

THE BOUNDARY MOVES

In continuous time the growth-decay boundary is $\mathrm{Re}\{a\} = 0$, the imaginary axis. In discrete time it is $|\alpha| = 1$, the unit circle. The map between them is $\alpha = e^{\beta}$.

1.13 When is a discrete-time exponential periodic?

This is the result that has no continuous-time counterpart. Require $x[n] = x[n+N]$ for $x[n] = Ce^{j\omega_0 n}$:

$$Ce^{j\omega_0 n} = Ce^{j\omega_0 (n+N)} \implies 1 = e^{j\omega_0 N} \implies j2\pi k = j\omega_0 N \implies N = \frac{2\pi}{\omega_0}k.$$

PERIODICITY CONDITION

$$\frac{\omega_0}{2\pi} = \frac{k}{N} \in \mathbb{Q}$$

N must be an **integer**. That is possible only when $\omega_0/2\pi$ is a rational number. If it is irrational, no integer N works and the sequence is aperiodic, no matter how sinusoidal it looks when plotted.

EXAMPLE 1.6

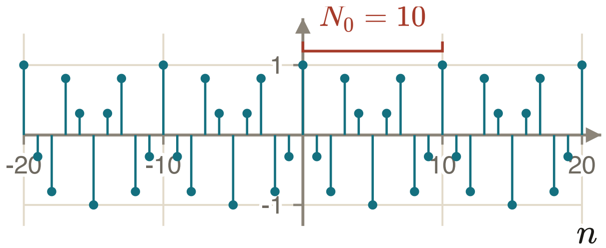
GIVEN $x[n] = e^{j(3\pi/5)n}$.

FIND The fundamental period N_0.

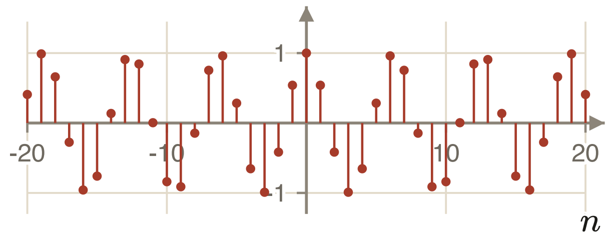
METHOD Use $N = 2\pi k/\omega_0$ and take the smallest k that makes N an integer.

SOLUTION $N = \dfrac{2\pi}{3\pi/5}k = \dfrac{10}{3}k$. The smallest positive integer k giving an integer N is $k = 3$, so $N_0 = 10$.

CHECK $\omega_0 N_0 = \dfrac{3\pi}{5} \times 10 = 6\pi = 2\pi \times 3$. The phasor makes exactly three full turns in ten samples, and k counts those turns.



$\mathrm{Re}\{e^{j3\pi n/5}\}$ repeats every 10 samples.



$\cos(n)$ has $\omega_0 = 1$, so $\omega_0/2\pi$ is irrational. It never repeats. It only looks as if it should.

> A SECOND DISCRETE-TIME FACT, NEEDED LATER
>
> $e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$ for every integer n. Discrete-time frequencies that differ by 2π cannot be told apart. The same fact makes the discrete-time Fourier transform 2π-periodic and makes aliasing unavoidable.

1.14 Chapter summary

1. E_∞ and P_∞ are limits. Finite energy forces zero average power. Finite non-zero average power forces infinite energy. Unbounded growth gives neither.
2. To build $x(at-b)$, shift by b first, then scale by a. The other order gives $x(at-ab)$.
3. T_0 and N_0 are the smallest positive periods, and $\omega_0 = 2\pi/T_0 = 2\pi/N_0$.
4. $\delta[n]$ is an ordinary sequence. $\delta(t)$ is a distribution defined by sifting.
5. Every signal is a sum of weighted, shifted impulses: $x[n] = \sum_k x[k]\delta[n-k]$.
6. A discrete-time exponential is periodic only when $\omega_0/2\pi$ is rational. A continuous-time one always is.

Exercises

1.1 Classify $x(t) = e^{-3t}u(t)$ as an energy signal, a power signal, or neither. Give the value of whichever quantity is finite.
Answer: Energy signal, $E_\infty = 1/6$.

1.2 For $x[n]$ non-zero only on $-1 \le n \le 1$ with values $1, 2, 1$, sketch $x[n+4]$ and $x[n-5]$ and give the support of each.
Answer: Supports $-5 \le n \le -3$ and $4 \le n \le 6$.

1.3 A signal $x(t)$ is non-zero only on $[-1, 5]$. On what interval is $x(2t+3)$ non-zero?
Answer: $[-2, 1]$.

1.4 Find the fundamental period of $x[n] = \cos(4\pi n/7)$, or show that it is aperiodic.
Answer: $N_0 = 7$.

1.5 Show that if $x(t)$ is odd then $\int_{-a}^{a} x(t)\,\mathrm{d}t = 0$ for every $a > 0$.

1.6 Evaluate $\int_{-\infty}^{\infty} (t^2+1)\,\delta(t-2)\,\mathrm{d}t$ and $\sum_{n=-\infty}^{\infty} 2^{-n}\,\delta[n-3]$.
Answer: 5 and $1/8$.

CHAPTER 2

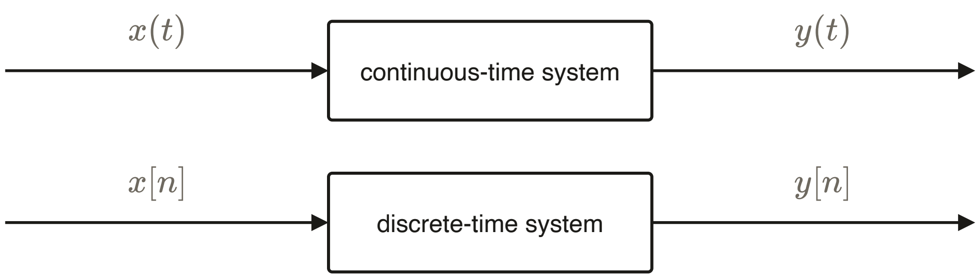
Systems and their properties

A system is judged by what it does to signals, not by what it is made of. Six properties decide almost everything that follows, and two of them are the whole of Chapter 3.

2.1 What a system is

A system is a rule that turns an input signal into an output signal. The rule is deterministic: the same input, applied twice, must give the same output. Without that, none of the tests below would even make sense.

Write it as an operator, $y = S\{x\}$. The argument is a whole function, not a number. That is why questions about memory, causality and time shifts can be asked at all.



The same picture in both domains. An amplifier, a numerical filter and a pencil-and-paper integrator can all be the same system.

2.2 Memory

CRITERION

A system is **memoryless** if the output at time t (or n) depends only on the input at that same time.

SYSTEM	VERDICT	REASON
$y(t) = \left[2x(t) - x^2(t)\right]^2$	memoryless	Only $x(t)$ appears. There is no $x(t+1)$ or $x(t-2)$ term.
$y[n] = x[n]$	memoryless	The identity system.
$y[n] = x[n-1]$	has memory	The output at n uses the sample at $n-1$.
$y[n] = x[n] + y[n-1]$	has memory	See the derivation below.

The last one is worth doing carefully, because the memory is hidden in the feedback. Substitute repeatedly:

$$y[n-1] = x[n-1] + y[n-2], \qquad y[n-2] = x[n-2] + y[n-3], \qquad \ldots$$

$$\implies \quad y[n] = x[n] + x[n-1] + x[n-2] + \cdots = \sum_{k=0}^{\infty} x[n-k].$$

The output depends on the whole past of the input. Feedback on the output is memory just as surely as an explicit delay on the input.

CIRCUIT READING

A resistor, $v(t) = R\,i(t)$, is memoryless. A capacitor, $v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,\mathrm{d}\tau$, is not. Memory is stored energy.

2.3 Invertibility

CRITERION

A system is **invertible** if distinct inputs always produce distinct outputs. The map must be one-to-one.

There are two ways to settle it, and they cost very different amounts of work. To prove invertibility, find an inversion formula that works for every input. To disprove it, find one pair of inputs that collide.

EXAMPLE 2.1

GIVEN $y(t) = \big[\cos(t) + 2\big]x(t)$.

FIND Is the system invertible?

SOLUTION Yes. The gain satisfies $1 \le \cos(t) + 2 \le 3$, so it never vanishes and

$$x(t) = \frac{y(t)}{\cos(t) + 2}.$$

CHECK Substituting back gives $\big[\cos t + 2\big]\dfrac{y(t)}{\cos t + 2} = y(t)$. Had the gain been $\cos(t)$ alone, the system would fail at every zero of the cosine.

EXAMPLE 2.2

GIVEN $y(t) = x^2(t)$.

FIND Is the system invertible?

SOLUTION No. Take $x_1(t) = 1$ and $x_2(t) = -1$ for all t. These are different inputs, but $y_1(t) = y_2(t) = 1$.

CHECK One counterexample is enough. The sign information is destroyed and cannot be recovered from y alone.

NOT AN INVERSION FORMULA

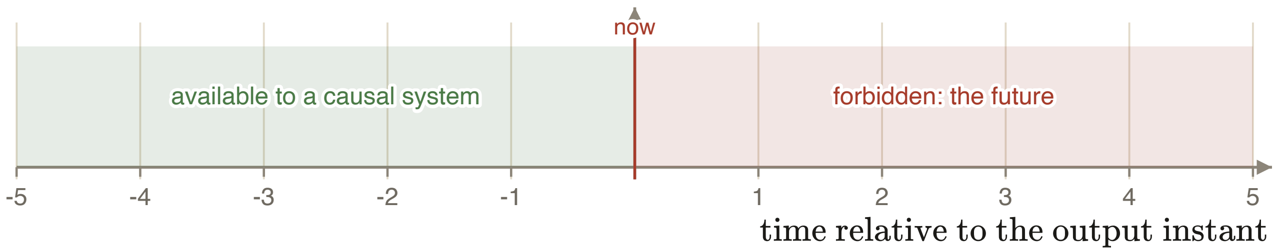
Writing $x(t) = \sqrt{y(t)}$ does not invert $y = x^2$. It picks one of two possible inputs by convention. An inversion formula must return the actual input, for every admissible input.

2.4 Causality

CRITERION

A system is **causal** if the output at time t (or n) depends only on inputs at times $\le t$: the present and the past.

SYSTEM	VERDICT	REASON
$y[n] = x[n-1]$	causal	Uses only a past sample.
$y[n] = x[n] + x[n+1]$	not causal	$x[n+1]$ is a future sample.
$y(t) = \int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau$	causal	The upper limit is t, so only $\tau \le t$ contributes.
$y[n] = x[-n]$	not causal	$y[-1] = x[1]$. The output at $n = -1$ needs a future input.
$y(t) = x(t)\cos(t+1)$	causal	Uses only $x(t)$. The factor $\cos(t+1)$ is a known function of t, fixed in advance, not a future input.



Causality restricts which part of the input axis the output is allowed to consult.

A system that must run in real time has to be causal, because the future has not happened yet. Non-causal systems are perfectly useful whenever the whole record already exists, as in offline audio or image processing.

2.5 Stability

A signal is **bounded** if there is a constant $B < \infty$ with $|x(t)| < B$ for all t.

CRITERION

A system is **BIBO stable** if every bounded input produces a bounded output. Bounded Input, Bounded Output.

THE TWO DIRECTIONS COST DIFFERENTLY

To prove stability you must bound the output for **every** bounded input. To disprove it you need **one** bounded input whose output is unbounded. Testing a single well-behaved input and concluding "stable" proves nothing.

EXAMPLE 2.3

GIVEN $y(t) = 2x^2(t-1) + x(3t)$.

FIND Is the system stable?

METHOD Assume $|x(t)| \le B < \infty$ and bound $|y|$ with the triangle inequality.

SOLUTION

$$|y(t)| \le \left|2x^2(t-1)\right| + \left|x(3t)\right| \le 2B^2 + B < \infty.$$

The system is stable.

CHECK Shifting and scaling the time axis cannot change the set of values a signal takes, so $x(t-1)$ and $x(3t)$ are bounded by the same B.

EXAMPLE 2.4

GIVEN $y[n] = \sum_{k=-\infty}^{n} x[k]$, the accumulator.

FIND Is the system stable?

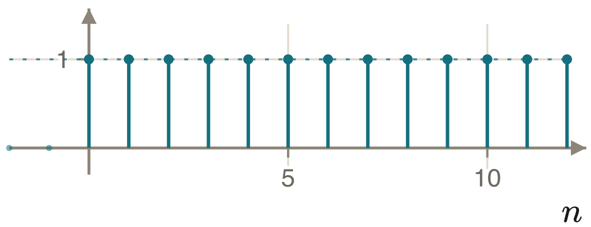
METHOD Look for one bounded input with an unbounded output.

SOLUTION Take $x[n] = u[n]$, so $|x[n]| \le 1$. Then

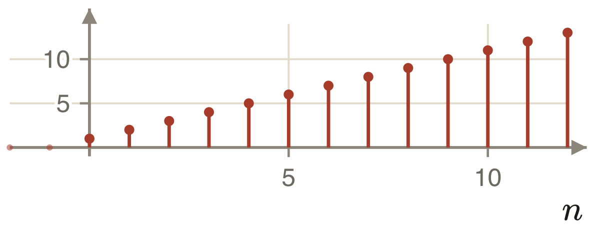
$$y[n] = \sum_{k=0}^{n} 1 = n + 1 \longrightarrow \infty.$$

The system is **not** stable.

CHECK The input never exceeds 1 and the output grows without bound. That is exactly the failure BIBO stability forbids.



Bounded input $u[n]$.



Unbounded output $n+1$.

CAUSAL DOES NOT MEAN STABLE

The accumulator is causal and unstable. The reversal $y[n] = x[-n]$ is stable and not causal. The six properties are independent except where a proof links them.

2.6 Time invariance

CRITERION

If $x(t) \to y(t)$, the system is **time invariant** when $x(t - t_0) \to y(t - t_0)$ for every t_0.

The test is a comparison of two computed signals.

1. **Path 1.** Shift the input, then run the system: $y_2 = S\{x(t - t_0)\}$.
2. **Path 2.** Run the system, then shift the output: $y_1(t - t_0)$.

The system is time invariant exactly when the two agree, for every input and every t_0.

EXAMPLE 2.5

GIVEN $y(t) = \sin\bigl(x(t)\bigr)$.

FIND Is the system time invariant?

SOLUTION Path 1: with $x_2(t) = x_1(t - t_0)$ we get $y_2(t) = \sin\bigl(x_1(t - t_0)\bigr)$.
Path 2: $y_1(t) = \sin\bigl(x_1(t)\bigr)$, so $y_1(t - t_0) = \sin\bigl(x_1(t - t_0)\bigr)$.
They agree for every t_0, so the system is time invariant.

CHECK The rule contains no explicit t. That is usually a reliable sign.

EXAMPLE 2.6

GIVEN $y[n] = n\,x[n]$.

FIND Is the system time invariant?

METHOD Look for a counterexample using an impulse.

SOLUTION Take $x_1[n] = \delta[n]$. Then $y_1[n] = n\,\delta[n] = 0$ for every n.
Now delay by one: $x_2[n] = \delta[n-1]$ gives $y_2[n] = n\,\delta[n-1] = \delta[n-1]$, which equals 1 at $n = 1$.
Time invariance would require $y_2[n] = y_1[n-1] = 0$. It does not. The system is **not** time invariant.

CHECK One counterexample settles it. Note that the impulse is the cheapest test signal available, because it isolates a single index.

THE PATTERN BEHIND EVERY FAILURE

A system fails time invariance exactly when its rule contains the time variable explicitly: a coefficient n, a gain $\cos(t)$, or a scaled argument $x(3t)$. If t or n appears anywhere outside the input, be suspicious at once.

2.7 Linearity

CRITERION

$$a\,x_1 + b\,x_2 \;\longrightarrow\; a\,y_1 + b\,y_2 \qquad \text{for all } a, b \in \mathbb{C}$$

where $x_1 \to y_1$ and $x_2 \to y_2$. This is additivity and homogeneity in one statement.

EXAMPLE 2.7

GIVEN $y(t) = 2\pi\,x(t)$.

FIND Is the system linear?

SOLUTION With $x_3 = ax_1 + bx_2$:

$$y_3 = 2\pi(ax_1 + bx_2) = a\underbrace{2\pi x_1}_{y_1} + b\underbrace{2\pi x_2}_{y_2} = ay_1 + by_2.$$

The system is linear. It is also time invariant, so it is the simplest possible LTI system.

EXAMPLE 2.8

GIVEN $y[n] = \big(x[2n]\big)^2$.

FIND Is the system linear?

SOLUTION With $x_3[n] = ax_1[n] + bx_2[n]$:

$$y_3[n] = \big(ax_1[2n] + bx_2[2n]\big)^2 = a^2x_1^2[2n] + 2ab\,x_1[2n]x_2[2n] + b^2x_2^2[2n],$$

while $ay_1[n] + by_2[n] = a\,x_1^2[2n] + b\,x_2^2[2n]$. The powers of a and b differ and a cross term appears, so the system is **not** linear.

CHECK A quicker route: scaling the input by a scales this output by a^2. Homogeneity alone already fails.

LINEAR DOES NOT MEAN STRAIGHT LINE

$y(t) = x(t) + 5$ is not linear, because the zero input does not give the zero output. Any linear system must satisfy $S\{0\} = 0$. That is a five-second first test.

2.8 How to classify an unfamiliar system

1. **Look for an explicit t or n.** A coefficient like n, a gain like $\cos t$, or a scaled argument like $x(3t)$ makes time invariance the first thing to test, and usually the first thing to fail.
2. **Look for nonlinearity.** Squares, products of the input with itself, $\sin(x)$, absolute values, saturation. Test homogeneity with a single scalar; it is the cheapest disproof available.
3. **Read the arguments.** Anything other than x evaluated exactly at t or n breaks memorylessness. Future arguments break causality.
4. **Try to bound the output.** Assume $|x| \le B$ and look for a finite bound. If the attempt fails, the failure usually shows you the counterexample.
5. **Leave invertibility to last.** It is needed least often and argued badly most often.

THE ONLY FREE IMPLICATION

Memoryless implies causal. Everything else must be established on its own. In particular, causal does not imply stable, linear does not imply time invariant, and stable does not imply memoryless.

SYSTEM	MEMORYLESS	INVERTIBLE	CAUSAL	STABLE	TIME INV.	LINEAR
$y(t) = 2\pi x(t)$	yes	yes	yes	yes	yes	yes
$y[n] = x[n-1]$	no	yes	yes	yes	yes	yes
$y(t) = x^2(t)$	yes	no	yes	yes	yes	no
$y[n] = n\,x[n]$	yes	no	yes	no	no	yes
$y[n] = \sum_{k\le n} x[k]$	no	yes	yes	no	yes	yes
$y[n] = x[-n]$	no	yes	no	yes	no	yes
$y[n] = (x[2n])^2$	no	no	no	yes	no	no

Read this table down the columns, not across the rows. No column is determined by another, which is why six separate tests are needed.

Exercises

2.1 Classify $y(t) = x(t)\,u(t)$ against all six properties.

Answer: Memoryless, not invertible, causal, stable, not time invariant, linear.

2.2 Show that $y[n] = x[n] - x[n-1]$ is invertible on inputs that are zero for $n < 0$, and give the inverse.

Answer: $x[n] = \sum_{k=0}^{n} y[k]$.

2.3 Give a system that is linear and causal but not stable, and one that is stable and time invariant but not linear.

2.4 Is $y(t) = x(t/2)$ time invariant? Prove or give a counterexample.

Answer: No. Path 1 gives $x(t/2 - t_0)$ and path 2 gives $x((t - t_0)/2)$.

CHAPTER 3

Linear time-invariant systems

Impose linearity and time invariance together, and a whole system is described by its response to a single impulse. This chapter derives that result and shows how to use it.

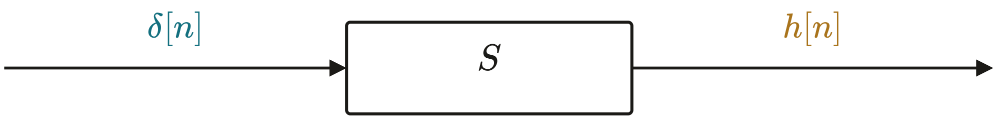
3.1 The impulse response

DEFINITION

The **impulse response** is the output when the input is a unit impulse: $x[n] = \delta[n]$ gives $y[n] = h[n]$. In continuous time, $x(t) = \delta(t)$ gives $y(t) = h(t)$.

For a general system this is one measurement among infinitely many. Knowing the response to $\delta[n]$ says nothing about the response to anything else.

For a linear time-invariant system it is everything. Time invariance turns one impulse response into a response for every shift. Linearity turns those into a response for every weighted sum. And Chapter 1 showed that every signal **is** a weighted sum of shifted impulses. Those three facts close the loop.



One experiment defines the whole system, provided the system is linear and time invariant.

3.2 Deriving the convolution sum

Start from the representation property of Chapter 1 and push it through the system.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] \quad \longrightarrow \quad S \quad \longrightarrow \quad y[n] = \,?$$

1. **Time invariance.** Since $\delta[n] \to h[n]$, we also have $\delta[n-k] \to h[n-k]$.
2. **Homogeneity.** The number $x[k]$ does not depend on n, so $x[k]\delta[n-k] \to x[k]h[n-k]$.
3. **Additivity.** The response to the sum is the sum of the responses.

CONVOLUTION SUM

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] \; = \; x[n] * h[n]$$

Substituting $m = n - k$ gives an equivalent form, which is already the statement that convolution is commutative:

$$\sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{m=-\infty}^{\infty} x[n-m]h[m].$$

THE PRECONDITION IS NOT OPTIONAL

Convolution applies only when the system is linear and time invariant. The derivation used time invariance once and linearity twice. Remove either property and no step survives. Applying convolution anyway produces a confident number that means nothing.

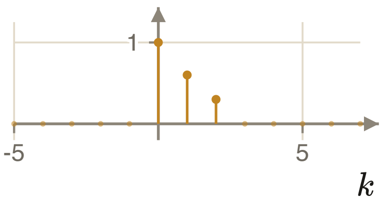
3.3 How to compute a convolution

1. **Flip.** Reverse $h[k]$ to get $h[-k]$.

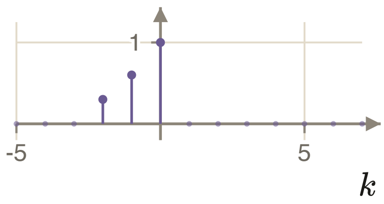
2. **Shift.** Move it by n to get $h[n-k]$.
3. **Multiply and add.** Form $x[k]h[n-k]$ and sum over k.
4. Repeat for every n.

Build $h[n-k]$ the same way you built $x(at-b)$ in Chapter 1. Treat k as the variable: shift $h[k]$ by n, then reverse in k.

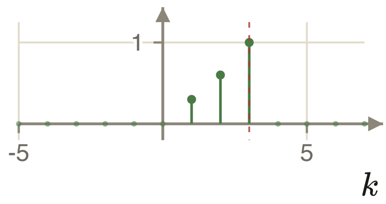
> WHAT HAPPENS IF YOU SKIP THE FLIP
>
> Sliding an unflipped h computes $\sum_k x[k]h[k-n]$, which is the cross-correlation of x and h, not the convolution. When h is symmetric the two agree, which is why this mistake often survives until an asymmetric h appears in an exam.



$h[k]$



$h[-k]$: flip



$h[n-k]$ at $n=3$: shift

EXAMPLE 3.1

GIVEN $x[n] = \{1, 2, 1, 2\}$ for $n = 0, 1, 2, 3$, and $h[n] = \{1, 1\}$ for $n = 0, 1$. Both zero elsewhere.

FIND $y[n] = x[n] * h[n]$.

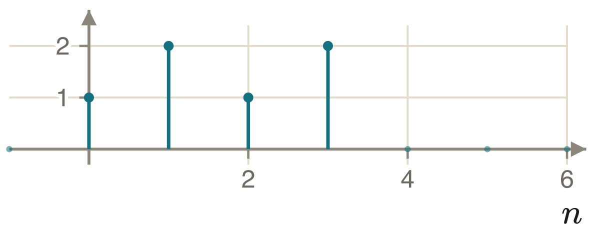
METHOD Expand the sum over the four non-zero samples of x and add the shifted copies of h.

SOLUTION

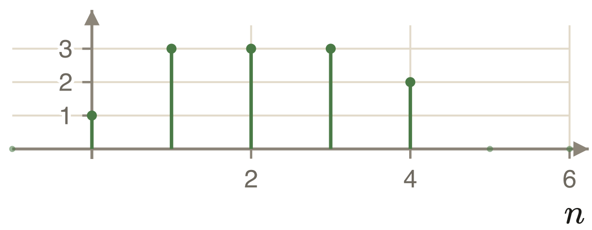
$$y[n] = x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + x[3]h[n-3] = h[n] + 2h[n-1] + h[n-2] + 2h[n-3]$$

Adding the four copies sample by sample gives $y = \{1, 3, 3, 3, 2\}$ on $n = 0, \ldots, 4$.

CHECK Two independent checks. **Length:** $4 + 2 - 1 = 5$ samples, and the support is $[0, 4]$. **Sum:** $\sum_n y[n]$ must equal $\left(\sum_n x[n]\right)\left(\sum_n h[n]\right)$, and $12 = 6 \times 2$. Also note that $h = \{1, 1\}$ is a two-point moving sum, so each output is the total of an adjacent input pair, which is what the numbers show.



$x[n]$



$y[n] = x[n] * h[n]$

EXAMPLE 3.2

GIVEN $x[n] = \left(\frac{1}{2}\right)^n u[n]$ and $h[n] = u[n]$.

FIND $y[n] = x[n] * h[n]$.

METHOD $h[n-k]$ equals 1 exactly when $k \le n$. Split on the sign of n.

SOLUTION **Case 1, $n < 0$.** The supports do not overlap: $x[k]$ needs $k \ge 0$ and $h[n-k]$ needs $k \le n < 0$. So $y[n] = 0$.
Case 2, $n \ge 0$. The overlap is $0 \le k \le n$, so

$$y[n] = \sum_{k=0}^{n} \left(\tfrac{1}{2}\right)^k = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 2 - \left(\tfrac{1}{2}\right)^n .$$

Together, $y[n] = \left(2 - \left(\frac{1}{2}\right)^n\right) u[n]$.

CHECK $y[0] = 1$, and directly $y[0] = x[0]h[0] = 1$. As $n \to \infty$, $y[n] \to 2$, which is the total sum of the input. That is what an accumulator should do.

GEOMETRIC SUM

$\sum_{k=m}^{n} a\, r^k = \dfrac{a\left(r^m - r^{n+1}\right)}{1-r}$. The finite sum needs only $r \ne 1$. The condition $|r| < 1$ is needed only when the sum runs to infinity.

3.4 The convolution integral

The continuous-time version follows the same three steps. Sifting gives the representation of x as a continuum of weighted impulses,

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\, \delta(t-\tau)\, \mathrm{d}\tau,$$

using $\delta(t) = \delta(-t)$. Pushing this through a linear time-invariant system gives:

CONVOLUTION INTEGRAL

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, \mathrm{d}\tau \;=\; x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, \mathrm{d}\tau$$

The two forms are equal. Flip whichever signal has the simpler shape.

BUILDING THE FLIPPED, SHIFTED H

Do it by shift, then scale, in the variable τ. For example, to plot $\delta(-t+5)$: first $v(t) = \delta(t+5)$, then $y(t) = v(-t) = \delta(-t+5)$, an impulse at $t = +5$, not at $t = -5$. Sign errors here are the main source of wrong integration limits.

EXAMPLE 3.3

GIVEN $x(t) = e^{2t}u(-t)$ and $h(t) = u(t-3)$.

FIND $y(t) = x(t) * h(t)$.

METHOD $x(\tau)$ is non-zero for $\tau \le 0$. And $h(t-\tau) = u(t-\tau-3)$ is non-zero for $\tau \le t-3$. So the overlap ends at $\min(0,\ t-3)$, and that is what creates two cases.

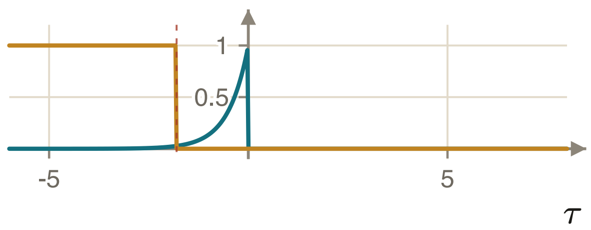
SOLUTION **Case 1, $t < 3$.** The binding limit is $t-3$:

$$y(t) = \int_{-\infty}^{t-3} e^{2\tau}\,\mathrm{d}\tau = \tfrac{1}{2}e^{2(t-3)}.$$

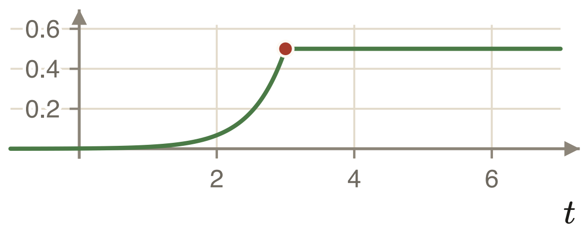
Case 2, $t > 3$. The binding limit is 0, so the whole of x is covered:

$$y(t) = \int_{-\infty}^{0} e^{2\tau}\,\mathrm{d}\tau = \tfrac{1}{2}.$$

CHECK Both branches give 0.5 at $t = 3$, so y is continuous. And $y(\infty)$ equals the total area of x, which is $1/2$. The system is a delayed integrator: it collects the area of x up to three seconds ago.



Case 1: $t < 3$. The shaded area is $y(t)$.



The result $y(t)$, continuous at $t = 3$.

EXAMPLE 3.4

GIVEN $x(t) = 1$ on $0 < t < 1$ and zero elsewhere. $h(t) = t$ on $0 < t < 2$ and zero elsewhere.

FIND $y(t) = x(t) * h(t)$.

METHOD Use $y(t) = \int h(\tau)x(t-\tau)\,\mathrm{d}\tau$ and flip the rectangle, which is the simpler shape. Its support becomes $[t-1,\ t]$: a window of width 1 sliding across the ramp on $[0, 2]$.

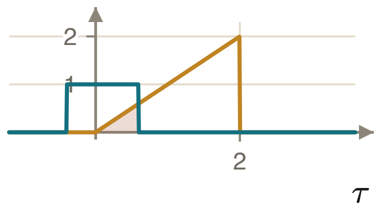
SOLUTION The window edges are $t-1$ and t. The ramp edges are 0 and 2. Setting them equal gives $t = 0, 1, 2, 3$: four boundaries, so five cases.

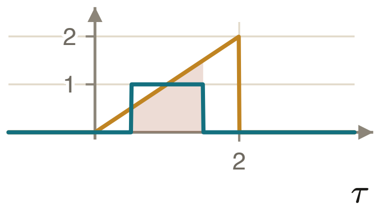
$t < 0$: no overlap, $y = 0$.
$0 < t < 1$: $y = \int_0^t \tau\,\mathrm{d}\tau = \tfrac{1}{2}t^2$.
$1 < t < 2$: $y = \int_{t-1}^t \tau\,\mathrm{d}\tau = t - \tfrac{1}{2}$.
$2 < t < 3$: $y = \int_{t-1}^2 \tau\,\mathrm{d}\tau = -\tfrac{1}{2}t^2 + t + \tfrac{3}{2}$.
$t > 3$: no overlap, $y = 0$.

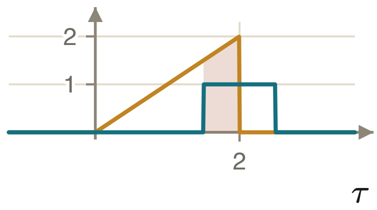
CHECK Three checks. **Continuity:** at $t = 1$ both branches give 0.5; at $t = 2$ both give 1.5; at $t = 3$ both give 0. **Support:** $[0, 1]$ and $[0, 2]$ give $[0, 3]$, because supports add. **Area:** $\int y = \left(\int x\right)\left(\int h\right) = 1 \times 2 = 2$, and $\frac{1}{6} + 1 + \frac{5}{6} = 2$.

FIND THE BOUNDARIES BEFORE INTEGRATING

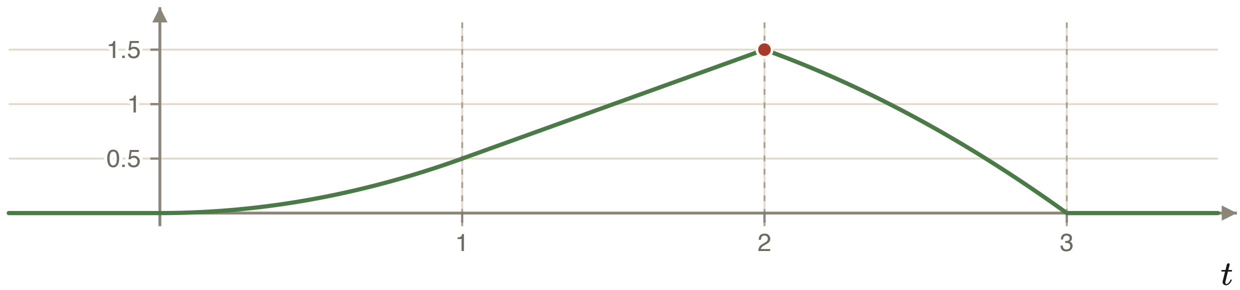
List the moving edges and the fixed edges, then set them equal. That produces the complete list of cases in one line, and no case can be lost by accident.







0 1 2



The complete result. The peak is 1.5 at $t = 2$. A unit-width rectangle acts as a one-second moving sum, so the output rises while the window fills, grows linearly while it is full, and falls as it leaves the ramp.

3.5 Properties of convolution

PROPERTY	STATEMENT	WHAT IT MEANS FOR INTERCONNECTIONS
Commutative	$x * h = h * x$	Input and impulse response play symmetric roles in the algebra.
Distributive	$x * (h_1 + h_2) = x * h_1 + x * h_2$	Two systems in **parallel**, outputs added, equal one system with impulse response $h_1 + h_2$.
Associative	$x * (h_1 * h_2) = (x * h_1) * h_2$	Two systems in **cascade** equal one system with impulse response $h_1 * h_2$.

Combining the last two with commutativity means the order of cascaded systems does not matter. That freedom is bought with linearity and time invariance, and it disappears the moment either is lost. A saturating amplifier followed by a filter is not the same system as the filter followed by the amplifier.

3.6 System properties in terms of h

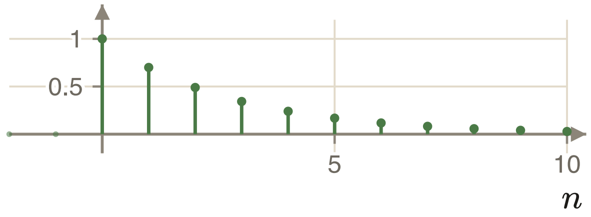
Every test of Chapter 2 becomes a statement about the single function h.

PROPERTY	CRITERION	REASON
Memoryless	$h(t) = a\,\delta(t)$, or $h[n] = a\,\delta[n]$	Then $y[n] = a\sum_k x[k]\delta[n-k] = a\,x[n]$, a pure gain.
Invertible	$h * g = \delta$ for some g	g is the impulse response of the inverse system, and δ is the identity system.
Causal	$h(t) = 0$ for $t < 0$, or $h[n] = 0$ for $n < 0$	The sum collapses to $y[n] = h[0]x[n] + h[1]x[n-1] + \cdots$, which uses only present and past inputs.
BIBO stable	$\sum_k \lvert h[k]\rvert < \infty$, or $\int \lvert h(t)\rvert\,\mathrm{d}t < \infty$	See the proof below.

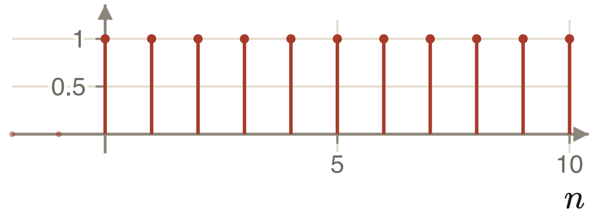
The stability condition is worth proving in one line. Assume $|x[n]| \le B$ for all n. Then

$$|y[n]| = \left|\sum_k h[k]x[n-k]\right| \;\le\; \sum_k |h[k]|\,|x[n-k]| \;\le\; B\sum_k |h[k]| < \infty.$$

This proves that absolute summability is **sufficient** for stability. It is also necessary; that direction is stated here without proof.



$h[n] = 0.7^n u[n]$. Here $\sum |h| = 1/0.3 < \infty$: stable and causal.



$h[n] = u[n]$, the accumulator. Here $\sum |h| \to \infty$: causal but not stable.

3.7 A convolution checklist

1. Confirm the system is linear and time invariant. Nothing below is valid otherwise.
2. Choose which signal to flip. Pick the simpler one.
3. Write the support of each factor as an inequality in the dummy variable.
4. Set the moving edges equal to the fixed edges to list every case boundary, before integrating anything.
5. Integrate or sum, case by case.
6. Check continuity at every boundary, check that the supports add, and check that the total area or total sum multiplies.

> WHAT THE THREE FINAL CHECKS CATCH
>
> A discontinuity at a boundary means a limit is wrong. A support that is too wide or too narrow means a flip or a shift is wrong. A total area that does not multiply means an integrand is wrong.

Exercises

3.1 Compute $\{1, 2, 3\} * \{1, -1\}$ with both sequences starting at $n = 0$, and check your answer with the sum rule.
Answer: $\{1, 1, 1, -3\}$; the sums are $6 \times 0 = 0$.

3.2 Let $h[n] = \left(\frac{1}{3}\right)^n u[n]$. Is the system stable? Is it causal? Is it memoryless?
Answer: Stable ($\sum |h| = 3/2$), causal, not memoryless.

3.3 Find $y(t) = u(t) * u(t)$ and sketch it.
Answer: $y(t) = t\,u(t)$.

3.4 Two systems with $h_1[n] = \delta[n] - \delta[n-1]$ and $h_2[n] = u[n]$ are cascaded. Find the impulse response of the cascade and identify the resulting system.
Answer: $\delta[n]$: the two systems are inverses.

3.5 $x(t) = 1$ on $0 < t < 3$ and $h(t) = t$ on $0 < t < 2$. List the case boundaries before computing anything, then find $y(t)$.
Answer: Boundaries at $t = 0, 2, 3, 5$; four cases.

APPENDIX A

Summary of formulas

A.1 Energy and power

QUANTITY	CONTINUOUS TIME	DISCRETE TIME
Total energy	$E_\infty = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2 \mathrm{d}t$	$E_\infty = \sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$
Average power	$P_\infty = \lim\limits_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \lvert x(t)\rvert^2 \mathrm{d}t$	$P_\infty = \lim\limits_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} \lvert x[n]\rvert^2$
Energy signal	$E_\infty < \infty$, $P_\infty = 0$	same
Power signal	$E_\infty \to \infty$, $0 < P_\infty < \infty$	same

A.2 Operations and periodicity

ITEM	STATEMENT
Shift	$x(t - t_0)$: delay if $t_0 > 0$, advance if $t_0 < 0$
Reversal	$x(-t)$, $x[-n]$
Scaling	$y(t) = x(at)$: compressed if $a > 1$, stretched if $0 < a < 1$; support divided by a
Combination	$x(at - b)$: shift by b first, then scale by a
Periodicity	$x(t) = x(t + T)$; $x[n] = x[n + N]$ with N an integer
Fundamental frequency	$\omega_0 = 2\pi/T_0 = 2\pi/N_0$
Even and odd	$\mathcal{E}\mathrm{v}\{x\} = \frac{1}{2}[x(t) + x(-t)]$, $\mathcal{O}\mathrm{d}\{x\} = \frac{1}{2}[x(t) - x(-t)]$

A.3 Impulses and steps

ITEM	CONTINUOUS TIME	DISCRETE TIME
Step and impulse	$\delta(t) = \frac{\mathrm{d}}{\mathrm{d}t}u(t)$, $u(t) = \int_{-\infty}^{t} \delta(\tau)\mathrm{d}\tau$	$\delta[n] = u[n] - u[n-1]$, $u[n] = \sum_{k=0}^{\infty} \delta[n-k]$
Sampling	$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$	$x[n]\delta[n - n_0] = x[n_0]\delta[n - n_0]$
Sifting	$x(t_0) = \int_{-\infty}^{\infty} x(t)\delta(t - t_0)\mathrm{d}t$	$x[n_0] = \sum_n x[n]\delta[n - n_0]$
Representation	$x(t) = \int x(\tau)\delta(t - \tau)\mathrm{d}\tau$	$x[n] = \sum_k x[k]\delta[n - k]$

A.4 Complex exponentials

ITEM	STATEMENT
Continuous time	$x(t) = Ce^{at}$; with $C = Ae^{j\theta}$ and $a = r + j\omega_0$: $x(t) = Ae^{rt}e^{j(\omega_0 t+\theta)}$
Period	$T_0 = 2\pi/\omega_0$, always periodic for $\omega_0 \neq 0$
Discrete time	$x[n] = C\alpha^n$ with $\alpha = e^{\beta}$; growth boundary at $\lvert\alpha\rvert = 1$
Period	$N = \frac{2\pi}{\omega_0}k$; periodic only if $\omega_0/2\pi \in \mathbb{Q}$
Frequency wrap-around	$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$

A.5 Systems and convolution

PROPERTY	GENERAL CRITERION	LTI CRITERION IN TERMS OF H
Memoryless	output at t uses only input at t	$h(t) = a\delta(t)$ or $h[n] = a\delta[n]$
Invertible	distinct inputs give distinct outputs	$h * g = \delta$ for some g
Causal	output uses only $\tau \leq t$	$h(t) = 0$ for $t < 0$; $h[n] = 0$ for $n < 0$
BIBO stable	bounded input gives bounded output	$\int \lvert h \rvert \, \mathrm{d}t < \infty$; $\sum_k \lvert h[k] \rvert < \infty$
Time invariant	$x(t - t_0) \to y(t - t_0)$	–
Linear	$ax_1 + bx_2 \to ay_1 + by_2$	–

CONVOLUTION

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,\mathrm{d}\tau$$

Commutative, distributive and associative. Supports add. Total areas or total sums multiply.

A.6 Symbols

SYMBOL	MEANING
$x(t)$, $x[n]$	input signal, continuous and discrete time
$y(t)$, $y[n]$	output signal
$h(t)$, $h[n]$	impulse response of a linear time-invariant system
$\delta(t)$, $\delta[n]$	unit impulse
$u(t)$, $u[n]$	unit step
E_∞, P_∞	total energy and average power over all time
T_0, N_0	fundamental period
ω_0	fundamental angular frequency, rad/s or rad/sample
$*$	convolution
j	imaginary unit, $j^2 = -1$